

# Digital Ecosystem-Based KPI-Driven Railway Communication Network Reporting System

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## ABSTRACT

This research is focused on architectural and modeling issues of design and development of digital reporting system aimed at the railway communication network infrastructure. Our approach to these problems is based on digital ecosystem paradigm and open-source Big Data technologies. It also aims at methodology for KPIs data preparation and collection in railway communication networks. A practical result of the research is a proposed software framework for digital reporting system that consists of multiple agents, which are important for efficient KPIs data integration and processing in communication network reporting system. We have created and tested a prototype digital reporting system using KPIs data from the production railway communication network. KPIs data used in test implementation and obtained from the largest Russian Railway company included all incidents, occurred in the railway communication network in a week period. Prototype system implementation successfully processed all KPIs data using proposed data methodology and integrated framework.

## CCS CONCEPTS

• Software and its engineering~Software organization and properties~Software system structures • Information systems~Information systems applications~Enterprise information systems~Enterprise applications

## KEYWORDS

Digital Ecosystems, Railway Communication Network Reporting, KPIs, Distributed Systems, Open-Source Big Data Technologies

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## 1 Introduction

Digitalization has become a priority at the micro and macro levels. Today, to remain competitive in the global market public corporations and companies from different sectors of the economy have to leverage their competencies in ICT taking advantage of new opportunities that digital transformation brings for new business models and demands for innovations in their respective ecosystems. These issues are in the focus of such national and international initiatives as Smart Manufacturing System (SMS) or Industries 4.0 (in the USA and Germany, respectively) [1].

The digital transformation of the railway communication network operations is contributing to an increase in data volumes across various functional units, by collecting and processing information from equipment, customers, and various business process management systems of the railway company. In this research, we suggest using an Open-Source Big Data techniques and services to handle data integration, storage and processing functions.

In this research, the Data Lake is suggested to be used as an enabling technology [2 – 9] for Digital Reporting system design. We put it in the core of architectural framework for the proposed Communication Network Reporting System (CNRS). The major services and stages of data processing pipeline for the real CNRS implementation in railway application domain are discussed in detail. Then, we model the KPIs data preparation process for operational and analytical KPIs in the Railway Communication Network. A methodology for KPI data organization in Railway communication network reporting is being developed. Finally, CNRS pipeline is modelled using real incidents data taken from a Railway Communication Network Company.

The proposed Communication Network Digital Reporting Ecosystem has been tested using Reporting KPIs data from one of the largest Railway Companies in Russia. The data consists of communication network incidents, which include seven entities. Data is used in our experiments for testing data processing workflow implementation and proposed methodology. Data volume was about 100 GB. The implementation of the pipeline has shown the successful data processing using proposed reporting system and methodology.

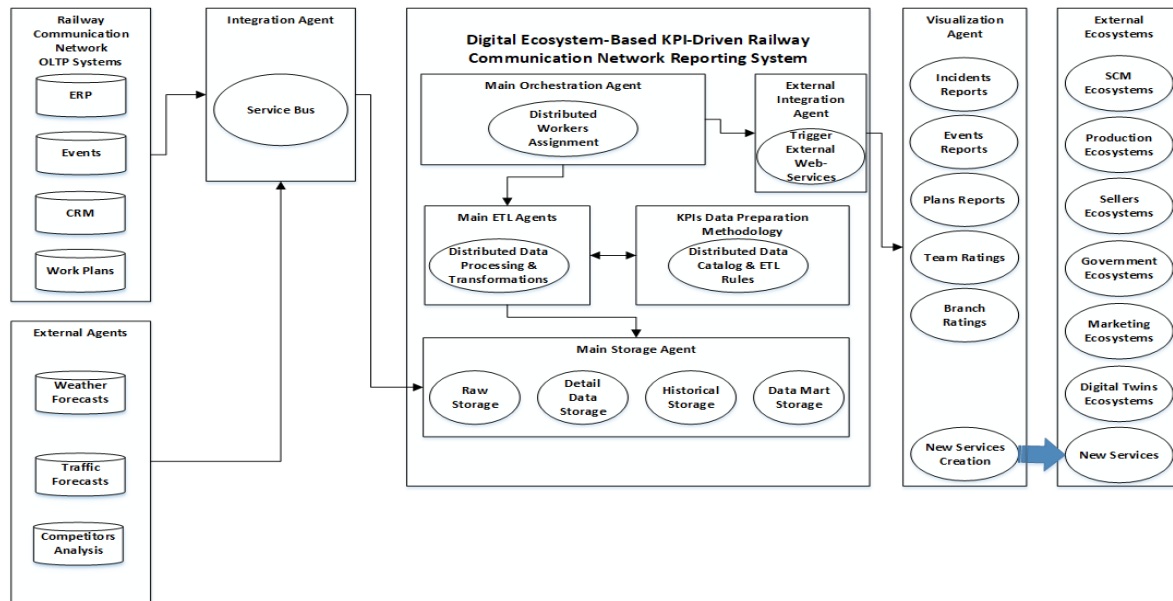


Figure 1: Architecture of Digital Ecosystem-Based KPI-Driven Railway Communication Network Reporting System.

## 2 Architecture of Railway CNRS

Figure 1 shows the functional architecture of the proposed Digital Ecosystem-Based KPI-Driven Railway Communication Network Reporting System.

It includes four major groups of services:

1. **Orchestration Main Service:** Service responsible for orchestrating all other services: ETL, Data Lake, External Integration service. It is the core part of the system, and all other integrations and new service designs will also be orchestrated using this service. It has a flexible distributed worker assignment algorithm, and only available, idle, and active worker nodes can assign tasks. The solution provides a robust, fault-tolerant, reliable, and scalable design for the ecosystem.
2. **External Integration Service:** A service that communicates with the BI service. The role of this service is to activate the BI service, and the loading of BI data should start immediately.
3. **Core ETL Services:** ETL data transformation and processing services across all data pipelines, right through to visualization. ETL can appear in any type of job, but should be distributed and fault-tolerant.
4. **Data Lake:** A service that stores all data in a distributed and reliable way such as historical Data, Detailed Data, and Data Mart. All data is stored in this service and all results of the processing are stored in this service. Data integrate with all source systems, data visualization for reporting. Based on visualized periodic reports, the administration will make decisions based on data, which will inevitably lead to the creation of new services and new ecosystems.

### 2.1 Main Agents Dataflow Description

Data can be traced from various consumer sources. Managing data is one of the most serious challenges faced by organizations today. Organizations are adopting the data lake models because they provide raw data that users can use for data experimentation and advanced analytics. A Data Lake could be a merging point of new and historic data, thereby drawing correlations across all data using advanced analytics. Data Lake can support self-service data practices to create added value from various data sources [11].

The data processing pipeline is strict and should go through the following Layers:

- **Staging Buffer Area:** The area where temporary data is accumulated in the source format without any changes.
- **Staging Exchange Area:** The region in between that will be used to create the following ETL processing packet. To build a data processing packet, all accumulated data is relocated from the buffer. A unique BATCH ID is assigned to it.
- **Operational Data Store:** The original data scheme is stored here, but, it is simplified to a single binary form of storage. It has a complete history of all changes and deletions.
- **Detail Data Store Staging:** Layering by batch for each source system, a unique instance is produced. Only between downloads is one-to-one data transferred from HDP and stored. It is possible to get a full data load as well as just line modifications (deltas).
- **Detail Data Store:** The current data slice's layer is displayed in a relational format. Putting together a single data model (without unification). Changing the keys (generation of internal storage IDs).

- **Data Mart:** The showcases of a group are organized by a specific attribute, most commonly the subject area. Contains a unified set of detailed information. It includes calculated indicators for reporting purposes. This layer receives calculations of indicators used in numerous reports.

## 2.2 Technologies for Railway Communication Network Reporting System Implementation

KPI-Driven Railway Communication Network Digital Reporting Ecosystem is built using Open-Source Big Data technologies, which satisfy following requirements:

- Horizontally scalable Agents for data storage – a layer for data loads to the archival and distributed storage.
- Agent with distributed data processing engine capable of processing large volumes of data.
- Fault-tolerant Agent – the solution should work without incurring data loss even in case of a single machine failure.
- All Agents should be orchestrated and combined in one main Agent that can run other Agents.
- Agent with a relational database for data access to different BI tools.
- Be Open Source.

Thus, we have defined the following Open-Source Big Data technologies for Railway Communication Network Digital Reporting Ecosystem implementation:

- Spark – a Distributed in-memory ecosystem for high-load data processing [12]. Spark has been used to create all main ETL Agents, which processed data and move it through all Agents;
- HDFS – a Distributed fault tolerant file system optimized for storage for processing large volumes of data [13]. A main Agent for distributed data storage of Railway data for Reporting in test implementation;
- PostgreSQL – a Relational database for universal data access from BI systems [14]. In test experiment, we used PostgreSQL as a main Agent for Serving layer and access to BI Agent;
- Airflow – an Orchestrator and Scheduler [15]. In our experiment, Airflow was main Agent for other Services orchestration and scheduling.

## 3 Methodology for KPIs Data Processing

From industrial process prospective, it is important that data collected in a specific industrial setup are linked with KPIs so that they can be analyzed to compare and measure with business strategy and organization. The KPI and other indicators can be used for generating composite indicators to be used in performance benchmarking with the best in industry, besides verifying the return on investments [16].

KPIs are used to determine a progress in achieving strategic and operational goals of a company, and to compare its performance with others within its industrial sector. Setting KPIs requires smart decision on how many indicators to track to determine the success of business. Industry-specific KPIs have been created in different markets including retail, healthcare, financial services, logistics, manufacturing, supply chain operations, and transportation [17].

Interviews with key users and business process owners on functional tasks, normative documentation of business process management, determined the basis of the developed solution combining formalized working principles and modern trends of building BI systems:

- A unified system of interrelated indicators. The use of cross-functional teams' methodologies for generating indicators makes it possible to create a unified information field for all participants at each level of interaction. The system of interlinked indicators, arranged in three levels of the hierarchy, includes more than 25 functional areas. Common rules for each level and area will provide trust in data and the development of a single vector of management decisions.
- A scenario-based approach to data analysis. Operational and analytical reports are designed in such a way that the user, once in the system, gradually progresses through a scenario from a general view of company performance to specific indicators of the functional area to solve the tasks at hand.
- The timeliness with which data can be retrieved by users. Industry specifics dictate the need to monitor performance indicators with a minimum time lag in order to respond to changes in a timely manner. A delay of even minutes in the decision-making process causes significant financial losses for the company, which cannot be ignored when designing the solution architecture.
- Data democratization - reporting available at all levels of the organizational structure. The role model of the system is designed so that top management, heads of business units as well as operational staff are involved in the process.
- A modern approach to building the UX/UI. The development of different scenarios for using the system from computers and video walls of different formats allowed to creation of a versatile and accessible solution, ready for scaling and development.

Fig. 2 shows the created Railway communication network methodology for operational and analytical KPIs.

## 4 Conclusion

In this paper, we have proposed a Digital Ecosystematic approach to the design and development of the KPI-Driven Railway Communication Network Reporting System that uses Big Data technologies to process KPI data efficiently, as well as a methodology for operational and analytical KPI data processing.

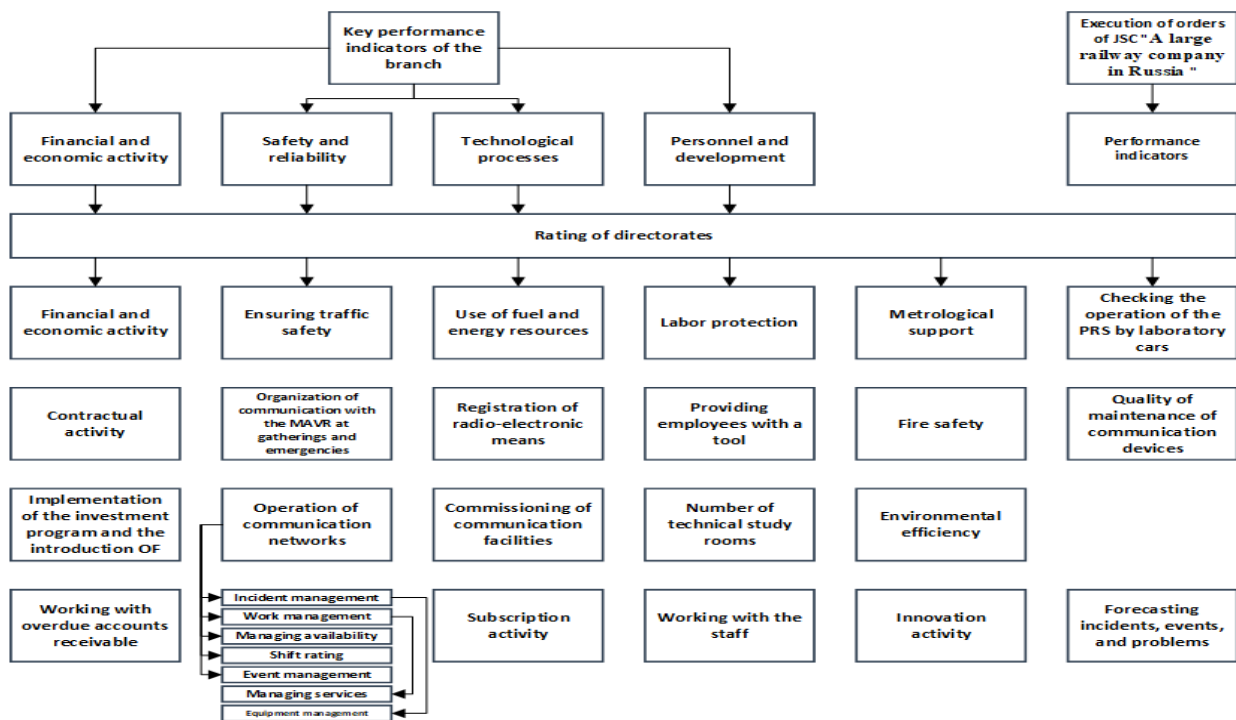


Figure 2: Railway communication network methodology for operational and analytical KPIs.

We reviewed the literature on data processing and storage for the Railway Communication Network and looked at data processing concerns in the Railway Communication Network Reporting sector, such as integration, orchestration, and BI subsystem integration. The Railway Communication Network Reporting system is based on the Digital Ecosystem paradigm, distributed services idea and Open-Source Big Data technologies. We suggest using Digital Ecosystem-based KPI-driven Communication Network Reporting System framework for Railway Communication Network Reporting, when high-loaded data pipelines are presented as well as a need to integrate many different data sources, orchestrate ETL-processes and structure KPIs of Railway Reporting using our methodology.

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